

Exploring Hallucination-Aware Extensions of HIRE for LVLMs

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Abstract

Base on hire, we try vti and interpretability

1. Introduction

Hallusination is important so there is a hire we extaned it

2. Related Work

Hallusiantion many studyed. Hire, VTI, Semnetic retain, CBM, Porotype

2.1. HIRE

Early methods rely on prompt filtering such as blacklisting or redaction, which are easily bypassed and fail to remove residual concept information due to contextual entanglement. MACE [?] addresses this by refining cross-attention and applying LoRA-based adaptation, but still requires re-training.

2.2. VTI

model training free but week sementic retain so projection

2.3. Prototpye

At vision classifier there is concept bottle netck to interpretability.

3. Method

Here we talk about what we do.

3.1. Sementic Representation in Training free

We try extend two way, traintig free hallusioanation. Make new tex at sec folder and input here

3.1.1. Problem Setup

use vti

3.1.2. Prompt Filtering via Subspace Projection

wkwjeifjakljdlkfa

3.2. Representation-level Hallucination Interpretability

Existing hallucination mitigation methods focus on reducing hallucinated outputs but provide limited interpretability of hallucination-related representations. To address this limitation, we introduce a prototype-based CBM that learns hallucination concepts in hidden representation space and analyzes representations through concept activations.

3.2.1. Prototype-based Concept Learning

CBMs require concept-level supervision, but hallucination concepts are not explicitly annotated in hidden representation space. To obtain interpretable concepts, we learn layer-wise prototypes corresponding to Safe and Existence hallucination categories and construct a prototype bank across all transformer layers.

Given a hidden representation h_i^l from transformer layer l , a prototype encoder projects it into a lower-dimensional embedding space z_i^l . The encoded representation is compared with learnable concept prototypes using cosine similarity $s_{i,l,c}$, where $c \in \{\text{safe}, \text{existence}\}$ denotes the concept category.

To obtain discriminative concept representations, we jointly optimize classification, clustering, and separation objectives. The classification loss encourages correct concept assignment, the cluster loss pulls samples toward the corresponding prototype, and the separation loss pushes samples away from other concept prototypes.

Algorithm 1 summarizes the prototype learning procedure.

3.2.2. Concept Bottleneck Module

After learning concept prototypes, we train a CBM. Instead of directly predicting hallucination labels from hidden representations, the CBM estimates concept activations from prototype similarities, providing an interpretable bottleneck between representations and hallucination prediction.

For each sample, prototype similarities from all transformer layers are aggregated into a concept vector v_i . The vector is processed by an MLP g_ϕ , and concept activations $\hat{c}_i$ are obtained through a sigmoid function.

Algorithm 1 Layer-wise Prototype Learning for Safe / Existence Concepts

Require: Layer-wise hidden representations $\mathcal{H} = \{(H_i, y_i)\}_{i=1}^N$, where $y_i \in \{\text{safe}, \text{existence}\}$

Require: $H_i = [h_i^1, h_i^2, \dots, h_i^L]$, where $h_i^l \in \mathbb{R}^{d_h}$

Require: Learnable layer-wise prototypes $\{p_{l,c}\}_{l=1, c \in \{\text{safe}, \text{exist}\}}^L$

Require: Prototype encoder $f_\theta : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_z}$

```

1: for each training epoch do
2:   for each mini-batch  $\mathcal{B} \subset \mathcal{H}$  do
3:     for each sample  $(H_i, y_i) \in \mathcal{B}$  do
4:       for each layer  $l = 1, \dots, L$  do
5:         Encode layer representation

```

$$z_i^l \leftarrow f_\theta(h_i^l)$$

```

6:       Normalize representation and proto-
       types

```

$$\tilde{z}_i^l \leftarrow \frac{z_i^l}{\|z_i^l\|_2}, \quad \tilde{p}_{l,c} \leftarrow \frac{p_{l,c}}{\|p_{l,c}\|_2}$$

```

7:       Compute prototype similarity

```

$$s_{i,l,c} = (\tilde{z}_i^l)^\top \tilde{p}_{l,c}$$

```

8:     end for
9:     Aggregate layer-wise similarities

```

$$\ell_{i,c} = \frac{1}{L} \sum_{l=1}^L s_{i,l,c}$$

```

10:    Compute prototype loss

```

$$\mathcal{L}_i = \mathcal{L}_{cls}^i + \lambda_c \mathcal{L}_{cluster}^i + \lambda_s \mathcal{L}_{sep}^i$$

```

11:  end for
12:  Compute mini-batch loss

```

$$\mathcal{L}_{proto} = \frac{1}{|\mathcal{B}|} \sum_{(H_i, y_i) \in \mathcal{B}} \mathcal{L}_i$$

```

13:  Update encoder and prototypes
14: end for
15: end for
16: return  $f_\theta, \{p_{l,c}\}$ 

```

To improve concept interpretability, we optimize the CBM using both prediction and concept supervision objectives. Binary cross-entropy encourages correct concept prediction, while sparsity regularization discourages multiple concepts from being activated simultaneously.

$$\mathcal{L}_{CBM} = \mathcal{L}_{CE} + \lambda_{BCE} \mathcal{L}_{BCE} + \lambda_{sparse} \mathcal{L}_{sparse}$$

Algorithm 2 Layer-wise Concept Bottleneck Training with Prototype Activations

Require: Layer-wise hidden representations $\mathcal{H}$

Require: Trained prototype bank $\{p_{l,c}\}$

Require: Prototype encoder f_θ

Require: Concept bottleneck layer g_ϕ

```

1: for each training epoch do
2:   for each mini-batch do
3:     for each sample do
4:       for each layer  $l$  do
5:         Encode representation

```

$$z_i^l = f_\theta(h_i^l)$$

```

6:       Compute similarity

```

$$s_{i,l,c} = \cos(z_i^l, p_{l,c})$$

```

7:     end for
8:     Aggregate concept vector

```

$$v_i = \left[\frac{1}{L} \sum_l s_{i,l,1}, \dots, \frac{1}{L} \sum_l s_{i,l,K} \right]$$

```

9:     Predict concept activations

```

$$\hat{c}_i = \sigma(g_\phi(v_i))$$

```

10:    Compute CBM loss
11:  end for
12:  Update CBM parameters
13: end for
14: end for
15: return  $g_\phi$ 

```

Algorithm 2 summarizes the CBM training procedure. The learned concept activations provide an interpretable view of hidden representations and enable hallucination analysis at the representation level. By preserving layer-wise prototype similarities, the proposed framework allows investigation of how hallucination-related information evolves.

4. Experiments

We test our project

4.1. Hallucination Interpretability

4.1.1. Experimental Setup

We evaluate the CBM on the generative subset of the AMBER dataset using LLaVA-1.5-7B and HIRE.

The dataset contains 9,848 Existence hallucination samples and 5,599 Safe samples. Hidden representations are extracted from all 32 transformer layers, and prototype sim-

Table 1. Hallucination evaluation on the COCO validation subset (100 images). Lower CHAIRs/CHAIRi is better, while higher Recall is better.

Model	CHAIRs ↓	CHAIRi ↓	Recall ↑
LLaVA	0.5200	0.1409	0.7940
HIRE	0.5000	0.1275	0.7209
LLaVA+CBM	0.5200	0.1417	0.7859
HIRE+CBM	0.5000	0.1276	0.7127

Table 2. Correlation between CBM Existence Metrics and CHAIR Scores

Metric	CHAIRi	CHAIRs
Avg. Existence Score	0.274	0.279
Max. Existence Score	0.291	0.330
Avg. Prototype Similarity	0.273	0.280
Max. Prototype Similarity	0.290	0.319

ilarity is measured using cosine similarity.

4.1.2. CBM Validation

Minimal Impact on Generation Table 1 reports CHAIR and Recall scores on the COCO validation subset. As expected, introducing the CBM module causes negligible changes in generation performance. The differences between the original models and their CBM counterparts remain minimal across all metrics. This is expected because the proposed framework is designed for representation analysis rather than hallucination mitigation.

Correlation with Hallucination Metrics To investigate whether the learned concepts are related to hallucination behavior, we compute correlations between CBM concept activations, prototype similarities, and CHAIR metrics. Table 2 presents the correlation coefficients. Although the correlations are moderate, all metrics exhibit positive relationships with both CHAIRi and CHAIRs. These observations suggest that the learned concepts capture hallucination-related characteristics in representation space.

4.1.3. Representation Analysis

Consistent Concept Separation To investigate whether the learned concepts are consistently separated, we analyze prototype margins and cross-layer prototype similarities.

As shown in Figure 1, Safe and Existence concepts begin to separate after approximately the third transformer layer, and the margin converges around the eighth layer. This indicates that the learned prototypes capture distinct representation patterns associated with hallucination behavior.

The cross-layer similarity matrix further shows strong diagonal patterns, suggesting that each layer learns a stable and distinct concept representation. Together, these results indicate consistent concept separation throughout the network.

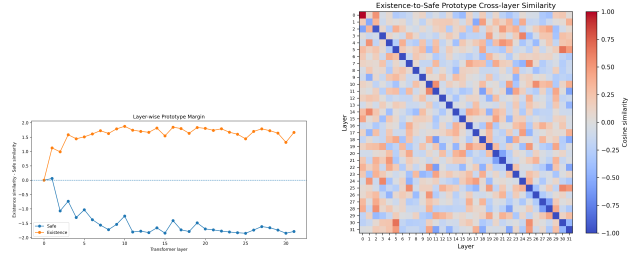


Figure 1. Concept separation analysis. Left: layer-wise prototype margin between Safe and Existence concepts. Right: cross-layer prototype similarity matrix.

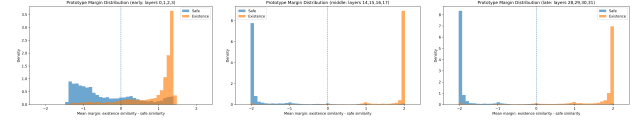


Figure 2. Prototype activation distributions across transformer layers.

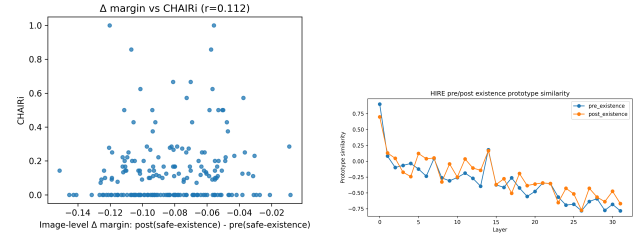


Figure 3. Limitations of the proposed framework. Left: relationship between concept activation changes and CHAIRi scores. Right: layer-wise Existence concept activations.

Stronger Signals in Deeper Layers We further analyze prototype activation distributions across early, middle, and late transformer layers.

Figure 2 shows substantial overlap between Safe and Existence concepts in early layers. However, the separation becomes increasingly clear in middle and deeper layers.

This trend suggests that hallucination-related information becomes progressively more distinguishable as representations move toward higher semantic abstraction levels.

4.1.4. Limitation

Although the proposed framework provides meaningful concept-level interpretations of hallucination-related representations, several limitations remain.

As shown in Figure 3, concept activations exhibit only moderate correlations with hallucination metrics. Furthermore, although Existence activations become increasingly pronounced in deeper layers, the learned concepts are not directly aligned with HIRE’s editing objective.

We believe this limitation arises because our framework focuses exclusively on Existence hallucinations, whereas HIRE addresses multiple hallucination categories simultaneously. Consequently, concept activations capture

Table 3. Inference overhead introduced by the layer-wise CBM module.

Model	Runtime (s)	With CBM (s)	Overhead
LLaVA	60	352	$5.87\times$
HIRE	139	1212	$8.72\times$

hallucination-related characteristics but remain weakly connected to actual hallucination mitigation performance.

The proposed framework also introduces additional computational overhead due to layer-wise representation extraction and prototype similarity computation.

As shown in Table 3, the additional analysis module increases inference time by approximately $5.9\times$ to $8.7\times$.

Future work will focus on extending the concept space beyond Existence hallucinations and integrating concept information directly into representation editing frameworks.

5. Conclusion

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References

- [1] Shilin Lu, Zilan Wang, Leyang Li, Yanzhu Liu, and Adams Wai-Kin Kong. Mace: Mass concept erasure in diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6430–6440, 2024.
- [2] Mengyao Lyu, Yuhong Yang, Haiwen Hong, Hui Chen, Xuan Jin, Yuan He, Hui Xue, Jungong Han, and Guiguang Ding. One-dimensional adapter to rule them all: Concepts diffusion models and erasing applications. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7559–7568, 2024.
- [3] Patrick Schramowski, Manuel Brack, Björn Deiseroth, and Kristian Kersting. Safe latent diffusion: Mitigating inappropriate degeneration in diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22522–22531, 2023.